

MARC AURÈLE GILLES

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RESEARCH RESUME

I develop rigorous, practical, and scalable algorithms for scientific data analysis, combining numerical linear algebra, statistics, and inverse problems. My current interests are heterogeneity analysis in cryogenic electron microscopy (cryo-EM) and randomized algorithms.

EDUCATION

Cornell University September 2014 - May 2019
Ph.D. in Applied Mathematics *Ithaca, NY*
Advised by Prof. Alex Townsend
Thesis title: At the intersection of differential equations and optimization: inverse problems, path planning and Krylov subspaces.

Rutgers University January 2012 - May 2014
B.A. in Mathematics *New Brunswick, NJ*
Summa Cum Laude
Minor in Economics

Raritan Valley Community College September 2010 - January 2012
Concentration in Economics *Branchburg, NJ*

PROFESSIONAL EXPERIENCE

Princeton University, Department of Mathematics September 2024 - Present
Assistant Professor of Mathematics *Princeton, NJ*

Princeton University, PACM September 2021 - September 2024
Postdoctoral Research Associate *Princeton, NJ*

- Advised by Prof. Amit Singer.
- Researched algorithms for reconstruction of protein structures by cryo-EM.
- Developed computational priors for Bayesian inference of protein structure and methods for covariance estimation of inferred potentials.
- Developed computational methods for heterogeneity analysis in cryo-EM.

Facebook Reality Labs July 2019 - September 2021
Research Scientist *Redmond, WA*

- Researched core technologies for augmented and virtual reality.
- Developed efficient algorithms for precise calibration, processing, and state estimation for novel sensors and imaging technologies.
- Invented and patented novel sensors for augmented reality.
- Led the optimal design of hardware components.
- Mentored and managed two Ph.D. interns on research projects.

Facebook Reality Labs

June 2018 - August 2018

*Research Intern**Redmond, WA*

- Designed novel computer vision and optimization algorithms for eye tracking and calibration.
- Implemented algorithms in C++ using OpenCV and Eigen.
- Worked alongside mechanical engineers and optical scientists to build experimental setups.
- Conceived and conducted user studies.

Argonne National Laboratory

June 2017 - August 2017

*Research Intern**Lemont, IL*

- Designed algorithms to perform 3D image reconstruction of nanometer-scale objects from X-ray measurements.
- Solved inverse problems with tens of millions of unknowns using supercomputers with tens of thousands of cores.
- Wrote high performance, massively parallel code in C using MPI and MKL.
- Collaborated with a team of physicists, engineers, and mathematicians.

**Center for Discrete Mathematics And
Theoretical Computer Science (DIMACS)**

June 2013 - August 2013

*Undergraduate Researcher**Piscataway, NJ*

- Developed computational imaging tools for biomedical applications under the supervision of Prof. Schliep.

TEACHING EXPERIENCE

Princeton University

January 2023 - Present

*Instructor**Princeton, NJ*

- (Fall 2026) - MAT 321: Numerical Analysis and Scientific Computing
- (Spring 2026) - Mathematics Junior Seminar: "Approximation Theory and Approximation Practice"
- (Spring 2026) - MAT 204: Advanced Linear Algebra with Applications
- (Fall 2025) - MAT 321/APC 321: Numerical Analysis and Scientific Computing
- (Spring 2025) - MAT 204: Advanced Linear Algebra with Applications
- (Fall 2024) - Mathematics Junior Seminar: The Top 10 Algorithms of the 20th Century
- (Fall 2024) - MAT 321/APC 321: Numerical Analysis and Scientific Computing
- (Spring 2024) - MAT 321/APC 321: Numerical Methods
- (Spring 2023) - MAT 321/APC 321: Numerical Methods

Cornell University

September 2014 - July 2019

*Teaching Assistant**Ithaca, NY*

- Prepared and held recitations
- Held office hours
- Wrote and graded exams.
- Classes taught:
 - INFO 2950 - Introduction to Data Science (Head TA)
 - CS4780/5780 - Machine Learning for Intelligent Systems
 - CS 1112 - Introduction to Computing Using MATLAB
 - MATH 1910 - Calculus for engineers
 - MATH 1920 - Multivariable Calculus for engineers
 - MATH 1110 - Calculus I

ADVISING

Postdoctoral researchers

- Yuguan Wang (co-advised with Amit Singer), Postdoctoral Research Associate, Princeton University (2026 - present). Extending RECOVAR to cryo-electron tomography

Graduate students

- Siji Chen, Ph.D. student, Program in Applied and Computational Mathematics, Princeton University (2025 - present)

Undergraduate students

- James Li, Princeton Undergraduate (Class of 2029). Summer research “Noise-robust variational autoencoders for cryo-EM heterogeneity analysis”
- Derek Wang, Princeton Undergraduate (B.A., Mathematics, 2028). Summer research on randomized numerical linear algebra
- Ryan Divan, Princeton Undergraduate (B.A., Mathematics, 2027). Research on randomly pivoted low-rank approximation; co-author of [1]
- Matias Andia, Princeton Undergraduate (B.A., Mathematics, 2027). Summer research “Tensor factorization methods for density estimation”
- Brandon Maher, Princeton Undergraduate (B.A., Mathematics, 2026). PACM certificate - project on “Linearly transformed sparse PPCA”
- Diego Rubio Garcia, Princeton Undergraduate (B.A., Mathematics, 2026). Senior thesis on randomized low-rank matrix completion for cryo-EM heterogeneity analysis
- Iniabasi Ekpenyong, Princeton Undergraduate (B.A., Computer Science, 2026). Summer research “Neural network-based solutions to high-dimensional PDEs”
- Won-Jae Chang (co-advised with Amit Singer). Princeton Undergraduate (B.A., economics, 2024). PACM certificate - project on “Efficient algorithms for Ewald sphere correction in cryo-EM”

HONORS AND PRIZES

- McGraw Center’s exemplar mentor award for graduate students and postdocs (2024)
- SIAM Student Travel Award (3×) in 2017 and 2018, 2023
- NSF Mathematical Sciences Graduate Internship in 2017
- Lawrence Corwin Memorial Math Prize in 2014
- Stanley E. Brasefield Mathematics Scholarship in 2013

FUNDING

- Co-PI, “RECOVAR-ET: Advancing Cryo-Electron Tomography Heterogeneity Analysis”, Chan Zuckerberg Biohub, \$200,000, July 2026 - June 2027 (PI: Amit Singer)

PUBLICATIONS AND PREPRINTS

1. Ryan Divan and Marc Aurèle Gilles. Convergence rates of randomly pivoted methods for low-rank approximation. *arXiv preprint arXiv:2609.06287*, 2026
2. Marc Aurèle Gilles. Convergence rates for pivoted QR and LU. *arXiv preprint arXiv:2607.26863*, 2026

3. Marc Aurèle Gilles and Eric J Verbeke. Computational problems from the 2026 3DEM Gordon Research Conference. Technical note, https://ma-gilles.github.io/notes/GRC_problems.pdf, 2026
4. Marc Aurèle Gilles and Heather Wilber. Low-rank approximation by randomly pivoted LU. *arXiv preprint arXiv:2601.22344*, 2026
5. Luke Evans, Lars Dingeldein, Roberto Covino, Marc Aurèle Gilles, Erik Thiede, and Pilar Cossio. Counting particles in cryo-electron microscopy may result in incorrect population estimates. *Communications Biology*, 9:421, 2026
6. Miro A Astore, Geoffrey Woollard, David Silva-Sánchez, Wenda Zhou, Mykhailo Kopylov, Khanh Dao Duc, Roy R Lederman, Yilai Li, Yi Zhou, Jing Yuan, Fei Ye, Quanquan Gu, Rémi Vuillemot, Slavica Jonic, Lan Dang, Steven J Ludtke, Hannah Bridges, Serena Liu, Michael McLean, Valentin Peretroukhin, Johannes Schwab, Eduardo R Cruz-Chú, Peter Schwander, Marc Aurèle Gilles, Amit Singer, David Herreros, Jose-Maria Carazo, Carlos O S Sorzano, J Ryan Feathers, Ellen D Zhong, Nikolaus Grigorie, Pilar Cossio, and Sonya M Hanson. The inaugural Flatiron Institute cryo-EM conformational heterogeneity challenge. *bioRxiv*, 2025
7. Marc Aurèle Gilles and Amit Singer. Cryo-EM heterogeneity analysis using regularized covariance estimation and kernel regression. *Proceedings of the National Academy of Sciences*, 122(9):e2419140122, 2025
8. Andy Zhang, Oscar Mickelin, Joe Kileel, Eric J Verbeke, Nicholas F Marshall, Marc Aurèle Gilles, and Amit Singer. Moment-based metrics for molecules computable from cryogenic electron microscopy images. *Biological Imaging*, 4:e3, 2024
9. Marc Aurèle Gilles and Amit Singer. A molecular prior distribution for bayesian inference based on wilson statistics. *Computer methods and programs in biomedicine*, 221:106830, 2022
10. Eric J Verbeke, Marc Aurèle Gilles, Tamir Bendory, and Amit Singer. Self fourier shell correlation: properties and application to cryo-et. *Communications Biology*, 7(1):101, 2024
11. Marc Aurèle Gilles and Alex Townsend. Continuous analogues of Krylov subspace methods for differential operators. *SIAM Journal on Numerical Analysis*, 57(2):899–924, 2019
12. Marc Aurèle Gilles, Christopher Earls, and David Bindel. A subspace pursuit method to infer refractivity in the marine atmospheric boundary layer. *IEEE Transactions on Geoscience and Remote Sensing*, 57(8):5606–5617, 2019
13. Marc Aurèle Gilles, Youssef Nashed, Ming Du, Chris Jacobsen, and Stefan Wild. 3D X-ray imaging of continuous objects beyond the depth of focus limit. *Optica*, 5(9):1078–1086, 2018
14. Marc Aurèle Gilles and Alexander Vladimirsky. Evasive path planning under surveillance uncertainty. *Dynamic Games and Applications*, 10(2):391–416, 2020

TALKS AND PRESENTATIONS

- Finding Latent Structure at Very Low SNR: Heterogeneity Analysis in Cryo-EM, Workshop on Mathematical Foundations of AI Models, Institute for Mathematical Sciences, National University of Singapore, Singapore, August 19, 2026
- What are the effects of the noise in heterogeneity analysis?, Three Dimensional Electron Microscopy Gordon Research Conference, Castelldefels, Barcelona, Spain, June 2026
- Reconstructing Biomolecular Heterogeneity from Cryo-EM Data (Poster), Curves and Surfaces, Saint-Malo, France, June 2026
- Low-Rank Approximation by Randomly Pivoted LU, Curves and Surfaces, Saint-Malo, France, June 9, 2026

- Low-Rank Approximation by Randomly Pivoted LU, 27th Conference of the International Linear Algebra Society (ILAS), Virginia Tech, Blacksburg, USA, May 20, 2026
- Low-Rank Approximation by Randomly Pivoted LU, Flatiron Institute, New York, USA, January 2026
- Reconstructing conformational states and inferring conformational densities in cryo-EM, 2025 Stanford SSRL/LCLS Users' Meeting, SLAC National Accelerator Laboratory, Menlo Park, California, 2025
- Reconstructing conformational states and inferring conformational densities in cryo-EM, Biophysical Society Annual Meetings, CryoEM Subgroup, Los Angeles, USA, 2025
- Reconstructing Conformational States & Densities in CryoEM with RECOVAR, OpenEye CUP XXIV, Santa Fe, USA, 2025
- Reconstructing Distributions of Molecules from Cryo-EM Datasets, PACM/CSML joint Colloquium, Princeton, USA, 2025
- Reconstructing Distributions of Molecules from Cryo-EM Datasets, Computational Microscopy Reunion Conference 2, Los Angeles, USA, 2025
- RECOVAR: A Bayesian framework for cryo-EM heterogeneity analysis (Selected poster talk), Gordon Research Conference, Barcelona, Spain, 2024
- RECOVAR: A Bayesian framework for cryo-EM heterogeneity analysis, Gordon Research Seminar, Barcelona, Spain, 2024
- Reconstructing flexible proteins from cryo-EM datasets, SIAM ALA, Sorbonne Université, 2024
- Reconstructing flexible proteins from massive microscopy datasets, Applied Mathematics Seminar, Yale University, New Haven, 2024
- Reconstructing flexible proteins from massive microscopy datasets, Applied Mathematics Seminar (online), University of Texas at Austin, 2024
- A Bayesian framework for cryo-EM heterogeneity analysis, CCM-CCB seminar, Flatiron Institute, NYC, 2023
- A Bayesian framework for cryo-EM heterogeneity analysis, Cryo-EM One World seminar series (online)
- Cryo-EM heterogeneity analysis by regularized covariance estimation, IAS, Gottingen, 2023
- Cryo-EM heterogeneity analysis by regularized covariance estimation, ICIAM, Tokyo, 2023
- Cryo-EM heterogeneity analysis by regularized covariance estimation, Cryo-EM summer workshop, Flatiron Institute, New York, 2023
- High Dimensional Covariance Estimation in Cryo-EM, SIAM MDS, San Diego, 2022
- Heterogeneity analysis in cryo-EM, IPAM seminar, Los Angeles, 2022
- Near Real-Time Heterogeneity Analysis by Sketched Covariance, GRC Three Dimensional Microscopy, Barcelona, Spain, 2022 (Selected Poster Presentation)
- A Molecular Prior Distribution for Bayesian Inference Based on Wilson Statistics, Cryo-EM seminar, Flatiron Institute, NY (online), 2022
- Computing with subspaces generated by differential operators, IDeAS seminar, Princeton University, 2021
- 3D X-ray imaging beyond the depth of focus limit, SIAM Conference on Imaging Science, 2018
- Continuous analogues of Krylov methods for differential operators, SIAM Conference on Applied Linear Algebra, 2018
- Continuous analogues of Krylov methods for differential operators, Scientific Computing and Numerics seminar, Cornell University, 2018
- Adversarial path planning, Scientific Computing and Numerics seminar, Cornell University, 2017
- A Subspace Pursuit Method to Invert the Refractivity Profile within the Marine Atmospheric Boundary Layer (Poster), SIAM Conference on Computational Science and Engineering, 2017

ORGANIZED EVENTS

- Co-organizer of the PACM IDeAS seminar, the weekly applied and computational mathematics seminar at Princeton University (Spring 2026 - present)
- Co-organized “Advances of regularization techniques in iterative reconstruction” minisymposium at SIAM Conference on Imaging Science (2018)
- Organized Applied Mathematics student-invited speaker series at Cornell University (2017-2019)

SOFTWARE

- RECOVAR [7]: <https://github.com/ma-gilles/recover>
- C++ companion library to [14]: github.com/eikonal-equation/Stationary_SEG
- MATLAB companion libraries to [11] : chebfun.org/examples/ode-linear/Krylov.html
- Python companion library https://github.com/ma-gilles/wilson_prior